Doping Stability Report -- Al in KCoO2 at 300 C

Generated: 2026-05-30 15:19   Host: KCoO2   Dopant: Al

Executive Summary

Al preferentially occupies the Co layer. The formation energy difference between the two layers is +6.309 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.00014 A^2/ps (stable), compared to -0.00308 A^2/ps on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-4.999	+0	No	PASS	PASS	STABLE
K	+1.310	+2	Yes	PASS	PASS	METASTABLE

Al at the Co layer -- STABLE

Charge situation

Al replaces Co with no change in oxidation state (isovalent substitution, charge mismatch = 0). No charge compensation is needed. The formation energy is fully reliable.

Formation Energy (thermodynamic stability)

Formation energy E_f	**-4.9992 eV**	PASS	*< 1.0 eV to pass*

E_f = -4.999 eV. This value is exceptionally large and negative, indicating that Al incorporation is strongly thermodynamically driven at the Co site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	**-0.00014 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.094 A^2**	----	
Max displacement	**0.746 A**	----	
Coordination (min/mean/max)	**4 / 4.0 / 5**	PASS	*relaxed = 5, tolerance +/-1*
Mean bond length (MD vs relaxed)	**1.781 A (relaxed: 1.8**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = -0.00014 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.094 A^2; maximum displacement from starting position = 0.75 A.

Coordination fluctuated between 4 and 5 (mean 4.0), within the +/-1 tolerance of the relaxed value (5). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 1.781 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Al is thermodynamically favorable and kinetically stable on the Co site at 300 C. All four stability criteria pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

Al at the K layer -- METASTABLE

Charge situation

Al replaces K with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+1.3102 eV**	FAIL	*< 1.0 eV to pass*

E_f = +1.310 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Al incorporation is thermodynamically unfavorable under equilibrium conditions at the K site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	**-0.00308 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.076 A^2**	----	
Max displacement	**0.647 A**	----	
Coordination (min/mean/max)	**4 / 4.0 / 5**	PASS	*relaxed = 5, tolerance +/-1*
Mean bond length (MD vs relaxed)	**1.797 A (relaxed: 1.8**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = -0.00308 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.076 A^2; maximum displacement from starting position = 0.65 A.

Coordination fluctuated between 4 and 5 (mean 4.0), within the +/-1 tolerance of the relaxed value (5). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 1.797 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: METASTABLE

Al sits stably on the K site during MD at 300 C, but the formation energy exceeds the 1.0 eV equilibrium threshold. This configuration is metastable -- it may be achievable via non-equilibrium synthesis routes such as ion exchange or rapid quenching.

Site Preference Comparison

The Co site is preferred by 6.309 eV over the K site. Formation energy: -4.999 eV (Co) vs +1.310 eV (K).

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Caveats and Limitations

? **CHGNet is charge-neutral.**

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch != 0) are approximate even when structural compensation defects are included.

? **No Freysoldt/Kumagai correction.**

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? **Metal-rich reference state.**

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? **MD timescales are short (25 ps default).**

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? **Energy rankings are more trustworthy than absolute values.**

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.